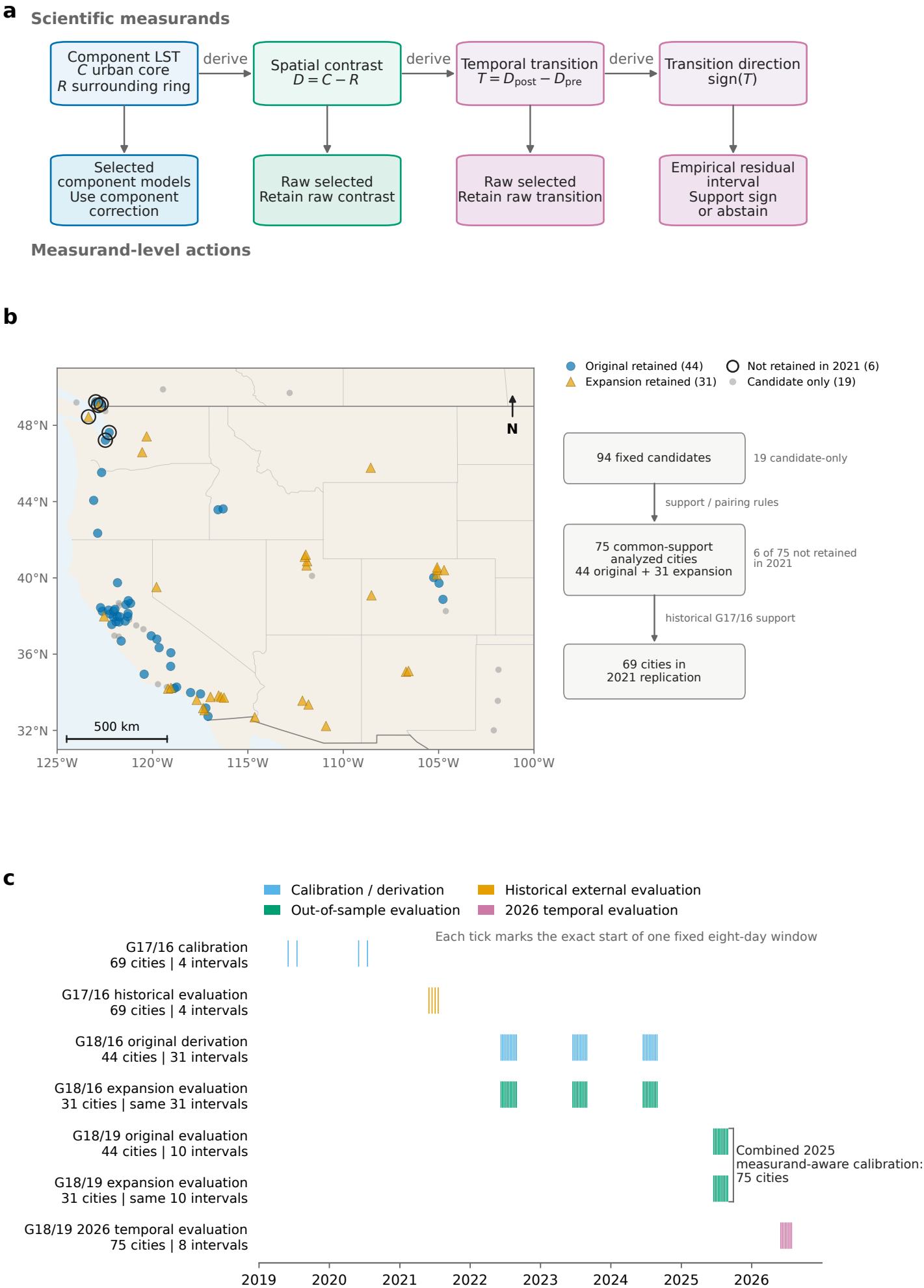
a
Scientific measurands
Component LST
C urban core
R surrounding ring
derive
Spatial contrast
D = C − R
derive
Temporal transition
T = D_post − D_pre
derive
Transition direction
sign(T)
Selected
component models
Use component
correction
Raw selected
Retain raw contrast
Raw selected
Retain raw transition
Empirical residual
interval
Support sign
or abstain
Measurand-level actions
b
Original retained (44)
Not retained in 2021 (6)
Expansion retained (31)
Candidate only (19)
N
48°N
44°N
40°N
36°N
32°N
500 km
125°W
120°W
115°W
110°W
105°W
100°W
94 fixed candidates
19 candidate-only
support / pairing rules
75 common-support
analyzed cities
44 original + 31 expansion
6 of 75 not retained
in 2021
historical G17/16 support
69 cities in
2021 replication
c
Calibration / derivation
Historical external evaluation
Out-of-sample evaluation
2026 temporal evaluation
Each tick marks the exact start of one fixed eight-day window
G17/16 calibration
69 cities | 4 intervals
G17/16 historical evaluation
69 cities | 4 intervals
G18/16 original derivation
44 cities | 31 intervals
G18/16 expansion evaluation
31 cities | same 31 intervals
G18/19 original evaluation
44 cities | 10 intervals
G18/19 expansion evaluation
31 cities | same 10 intervals
Combined 2025
measurand-aware calibration:
75 cities
G18/19 2026 temporal evaluation
75 cities | 8 intervals
2019
2020
2021
2022
2023
2024
2025
2026